

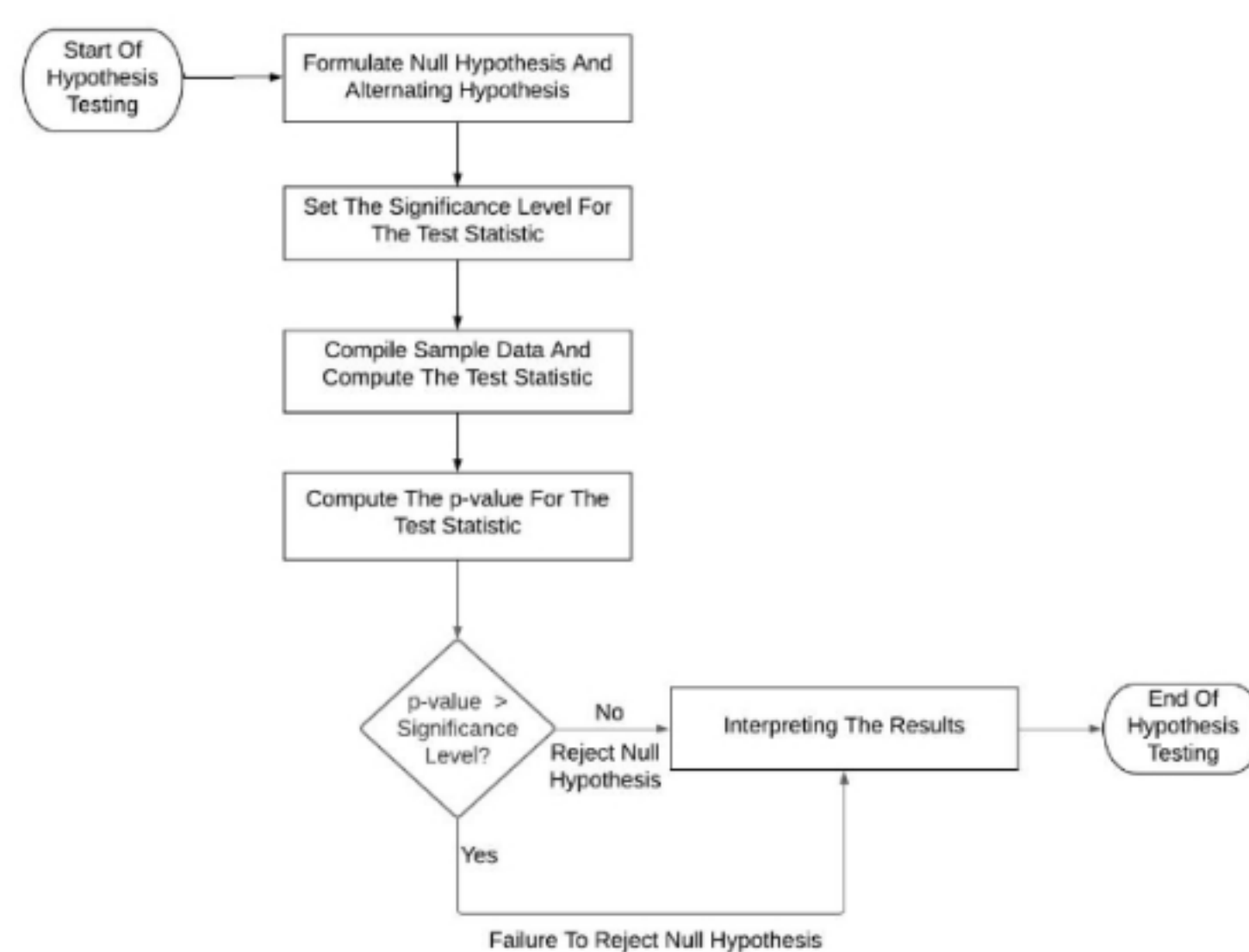


What is Hypothesis Testing?
Explain Null and Alternative
Hypothesis

with examples in Data science



What is Hypothesis Testing?



Null vs. Alternative

Null Hypothesis

H_0

A statement about a population parameter.

We test the likelihood of this statement being true in order to decide whether to accept or reject our alternative hypothesis.

Can include =, ≤, or ≥ sign.

ThoughtCo.

Hypothesis Testing is a statistical method used in **data science** to make decisions or inferences about a population based on sample data.



In simple terms:



It helps us decide whether a claim

(assumption) about data is true or false

us



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Hypothesis Testing is a statistical method used in **data science** to make decisions or inferences about a population based on sample data.



In simple terms:

It helps us decide whether a claim (assumption) about data is **true or false** using evidence.



Key Idea

We start with an assumption and then test it using data.



1. Null Hypothesis (H_0)



The **Null Hypothesis (H_0)** is the **default assumption** or **status quo**.

It states that **there is no effect, no difference, or no relationship**.



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1. Null Hypothesis (H_0)



The **Null Hypothesis (H_0)** is the **default assumption** or **status quo**.

It states that **there is no effect, no difference, or no relationship**.



It is assumed **true initially** until proven otherwise.



Examples:

1. In a medical study:

H_0 : A new drug has **no effect** on patients.

2. In a website test (A/B testing):

H_0 : New design does **not increase** user engagement.

3. In a machine learning model:

H_0 : Model accuracy is **equal to baseline accuracy**.



2. Alternative Hypothesis (H_1 or H_a)



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The **Alternative Hypothesis (H_1)** is the **opposite of H_0** .

It states that **there is an effect, difference, or relationship**.



2. Alternative Hypothesis (H_1 or H_a)



The **Alternative Hypothesis (H_1)** is the **opposite of H_0** .

It states that **there is an effect, difference, or relationship**.



This is what we **want to prove**.



Examples:

1. Medical study:

H_1 : The new drug **improves patient health**.

2. Website testing:

H_1 : New design **increases user engagement**.

3. ML model:

H_1 : Model accuracy is **better than baseline**.



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Decision Rule

After testing:

- If evidence is strong → **Reject H_0**
- If evidence is weak → **Fail to reject H_0**

⚠ Note: We never “accept” H_0 , we only say we don’t have enough evidence to reject it.



Types of Alternative Hypothesis

1. Two-tailed test

H_1 : Parameter $\neq$ value
(e.g., mean $\neq$ 50)

2. One-tailed test

- Right-tailed: Parameter $>$ value
- Left-tailed: Parameter $<$ value



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Example



Real-Life Data Science Example



Example: E-commerce Conversion Rate

- H_0 : New checkout system does **not** change conversion rate
- H_1 : New checkout system **improves** conversion rate



You collect user data and compute a **p-value**:

- If $p\text{-value} < 0.05 \rightarrow$ Reject $H_0 \rightarrow$ New system works
- If $p\text{-value} \geq 0.05 \rightarrow$ Fail to reject H_0



Why Hypothesis Testing is Important in Data Science



- Validates assumptions using data

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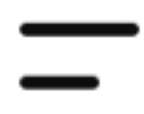


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nd

- Helps avoid decisions based on



You collect user data and compute



p-value:

- If $p\text{-value} < 0.05 \rightarrow \text{Reject } H_0 \rightarrow \text{New system works}$
- If $p\text{-value} \geq 0.05 \rightarrow \text{Fail to reject } H_0$

Why Hypothesis Testing is Important in Data Science

- Validates assumptions using data
- Used in **A/B testing, model evaluation, and experiments**
- Helps avoid decisions based on guesswork

Short Exam Definition (Write This)

Hypothesis Testing ↓ Statistical method used to determine whether there is enough evidence to reject the null hypothesis.



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The **Null Hypothesis** (H_0) represents no effect or no difference, while the **Alternative**



Differentiate between One-Way and Two-Way Hypothesis Tests with examples.

Here's an **exam-oriented answer** (7–8 marks) with clear differentiation and examples:

Difference Between One-Way and Two-Way Hypothesis Tests

1. Definition

- **One-Way Hypothesis Test (One-Tailed Test):**

A test where the **alternative hypothesis checks for a difference in only one direction** (either greater than or less than).



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A test where the **alternative hypothesis checks for a difference in both directions**



Difference Between One-Way and Two-Way Hypothesis Tests

1. Definition

- **One-Way Hypothesis Test (One-Tailed Test):**
A test where the **alternative hypothesis checks for a difference in only one direction** (either greater than or less than).
- **Two-Way Hypothesis Test (Two-Tailed Test):**
A test where the **alternative hypothesis checks for a difference in both directions** (either greater than or less than).

2. Key Differences

Basis

One-Way Test

Direction



Only one direction



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(H_1)

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2. Key Differences

Basis	One-Way Test
Direction	Only one direction ↓)
Alternative Hypothesis (H_1)	$\mu > \mu_0$ or $\mu < \mu_0$
Rejection Region	Only one tail of distribution
Critical Region	Entire α in one side
Sensitivity	More sensitive in direction
Usage	When direction is known

3. Graphical Representation



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One-tailed Test

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2. Key Differences

One-Way Test

	Only one direction ($\uparrow$ or $\downarrow$)
Null Hypothesis	$\mu > \mu_0$ or $\mu < \mu_0$
Region	Only one tail of distribution
Significance level	Entire α in one side
Power	More sensitive in one direction
When to use	When direction is known

3. Graphical Representation



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One-tailed Test



2. Key Differences

Test	Two-Way Test
direction ($\uparrow$ or $\downarrow$)	Both directions ($\uparrow$ and $\downarrow$)
$\mu < \mu_0$	$\mu \neq \mu_0$
tail of distribution	Both tails of distribution
α on one side	α divided into two ($\alpha/2$ each side)
sensitive in one direction	Detects any difference
Direction is known	When direction is unknown

3. Graphical Representation



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One-tailed Test



3. Graphical Representation

One-Tailed Test

- Rejection region lies **only on one side**

Example:

- Right-tailed: $\mu > \mu_0$
 - Left-tailed: $\mu < \mu_0$
-

Two-Tailed Test

- Rejection region lies **on both sides**

Example:

- $\mu \neq \mu_0$
-



4. Examples in Data Science

(A)



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Problem:

A company claims that average user session



4. Examples in Data Science

(A) One-Way Test Example

Problem:

A company claims that average user session time is **greater than 5 minutes**.

- $H_0: \mu \leq 5$
- $H_1: \mu > 5 \rightarrow$ (Right-tailed test)

Use Case:

Checking if a **new feature increases engagement**

(B) Two-Way Test Example

Problem:

A website redesign may **change** user session time.

- $H_0: \mu = 5$
- $H_1: \mu \neq 5 \rightarrow$ (Two-tailed test)



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Checking if a new design affects behavior
(increase or decrease)



(B) Two-Way Test Example

Problem:

A website redesign may **change** user session time.

- $H_0: \mu = 5$
- $H_1: \mu \neq 5 \rightarrow (Two\text{-}tailed\ test)$

Use Case:

Checking if a **new design affects behavior** (increase or decrease)

5. When to Use

- Use **One-Way Test** when:
 - You are interested in **only one direction**
 - Example: Increase in sales, decrease in error rate
- Use **Two-Way Test** when:
 - You want to detect **any change**



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- You are interested in **only one direction**
- Example: Increase in sales, decrease in error rate
- Use **Two-Way Test** when:
 - You want to detect **any change**
 - Example: Change in model accuracy, change in user behavior

6. Conclusion

- **One-way test** focuses on **specific direction** and is more powerful in that direction.
- **Two-way test** is more **general and unbiased**, used when direction is not known.



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5-n



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Explain Statistical Significance, P-value, Type I Error, and Type II Error.

Here's a **clear, exam-oriented explanation** of all four concepts:



1. Statistical Significance

Definition:

Statistical significance tells us whether the result obtained from data is **real (not due to chance)**.

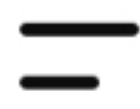
- It is determined using a **significance level (α)**, usually:
 - 0.05 (5%) or 0.01 (1%)

Interpretation:

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- If not → Fail to reject H_0



1. Statistical Significance

Definition:

Statistical significance tells us whether the result obtained from data is **real (not due to chance)**.

- It is determined using a **significance level (α)**, usually:
 - 0.05 (5%) or 0.01 (1%)

Interpretation:

- If result is **statistically significant** → Reject Null Hypothesis (H_0)
- If not → Fail to reject H_0

Example (Data Science):

A model shows 90% accuracy. If the result is statistically significant at $\alpha = 0.05$, it means the performance is **real and not random**.



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2. P-value

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2. P-value

Definition:

The **p-value** is the probability of obtaining the observed result **assuming the Null Hypothesis (H_0)** is true.

Decision Rule:

- If **p-value $\leq \alpha$** $\rightarrow$ Reject H_0
- If **p-value $> \alpha$** $\rightarrow$ Do not reject H_0

Interpretation:

- Small p-value $\rightarrow$ Strong evidence against H_0
- Large p-value $\rightarrow$ Weak evidence against H_0

Example:

- p-value = 0.03, $\alpha = 0.05$
- Since $0.03 < 0.05 \rightarrow$ Reject $H_0 \rightarrow$ Result is significant



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3. Type I Error (False Positive)

⚠️ 3. Type I Error (False Positive)

Definition:

Rejecting the Null Hypothesis when it is actually **true**.

👉 You think something exists, but it actually doesn't.

Probability of Type I Error = α

Example:

- A test says a drug works (reject H_0)
- But in reality, it doesn't work
→ Type I Error

⚠️ 4. Type II Error (False Negative) ↓

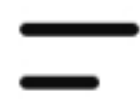
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actually false.



⚠️ 4. Type II Error (False Negative)

Definition:

Failing to reject the Null Hypothesis when it is actually **false**.

👉 You think nothing is happening, but actually it is.

Probability of Type II Error = β

Example:

- A test says a drug does not work
- But in reality, it works
→ Type II Error



Summary Table (Very Important for Exams)



C/



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Statistical Significance Result is not due to chance



What are t-Tests? Explain types of t-tests and degrees of freedom.

t-Tests (Student's t-test)

A **t-test** is a statistical hypothesis test used to determine whether there is a **significant difference between means** of one or two groups when:

- Sample size is **small** ($n < 30$)
- Population **standard deviation is unknown**
- Data is approximately **normally distributed**

It was developed by William Sealy Gosset (under the pseudonym "Student").

Types of t-Tests

1. One-Sample t-Test



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Use

sample with a known or hypothesized population mean.

It was developed by William Sealy Gosset (under the pseudonym "Student").



Types of t-Tests

1. One-Sample t-Test

Used to compare the **mean of a single sample** with a known or hypothesized population mean.

- **Example:**

Checking whether the average marks of a class differ from 60.

- **Formula:** $t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$

Where:

- $\bar{x}$ = sample mean
- μ = population mean
- s = sample standard deviation
- n = sample size



2.



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Used to compare the means of two independent groups



2. Independent (Two-Sample) t-Test

Used to compare the **means of two independent groups**.

- **Example:**

Comparing average salary of employees in two different companies.

- **Types:**

- Equal variance (pooled t-test)
- Unequal variance (Welch's t-test)

- **Key idea:**

Tests whether the difference between two means is statistically significant.

3. Paired t-Test (Dependent t-Test)

Used when observations are **related or paired**.



- **Example:**



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- **Formula:**

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

=



3. Paired t-Test (Dependent t-Test)

Used when observations are **related** or **paired**.

- **Example:**

Before-and-after test scores of the same students.

- **Formula:** $t = \frac{\bar{d}}{s_d / \sqrt{n}}$

Where:

- $\bar{d}$ = mean of differences
- s_d = standard deviation of differences
- n = number of pairs

Degrees of Freedom (df)



Degrees of freedom represent the number of independent observations.



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- s_d = standard deviation of differences

= : number of pairs



Degrees of Freedom (df)

Degrees of freedom represent the number of independent values that can vary in a statistical calculation.

Formulas for Different t-Tests

1. One-Sample t-Test:

$$df = n - 1$$

2. Independent t-Test:

- Equal variance:

$$df = n_1 + n_2 - 2$$

Uses a complex formula
(approximate df)



3. Paired t-Test:



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es a complex formula
(approximate df)



3. Paired t-Test:

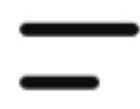
$$df = n - 1$$

Key Points

- t-tests are used when **variance is unknown and sample size is small**
- Based on **t-distribution**, which is wider than normal distribution
- As sample size increases, t-distribution approaches **normal distribution**
- Used widely in **data science, research, and experiments**

Conclusion (Ex ↓ Oriented)

A **t-test** is a statistical method used to test
by **+** Reply to ChatGPT  
the population variance is unknown. The
main types are **one-sample, independent,**



What is ANOVA? Explain
One-Way ANOVA and
F-statistics with
examples



What is ANOVA?

ANOVA (Analysis of Variance) is a statistical technique used to compare the **means of three or more groups** to determine whether at least one group mean is significantly different from the others.

👉 Instead of comparing means directly, ANOVA compares **variances**:

- **Between-group variance** (variation due to treatment or groups)
- **Within-group variance** (variation due to random error)



Purpose:

To test the null hypothesis:



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$H_0 : \mu_1 = \mu_2 = \mu_3 = \dots$

=



• **within-group variance** (variation due to random error)

Purpose:

To test the null hypothesis:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \dots$$

One-Way ANOVA

One-Way ANOVA is used when:

- There is **one independent variable (factor)**
- That factor has **3 or more levels (groups)**

◆ Example:

Suppose we want to test if **different teaching methods** affect student scores:

Method A	Method B	Method C
70	65	80

75



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72

70

78

Example:

=



Suppose we want to test if **different teaching methods** affect student scores:

Method A	Method B	Method C
70	65	80
75	68	85
72	70	78

👉 Here:

- Factor: Teaching Method
- Groups: A, B, C
- Dependent variable: Scores

◆ Steps in One-Way ANOVA:

1. State hypotheses:

- H_0 : All mean equal
- H_1 : At least one mean is different

2

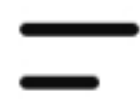


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◦ Group means

◦ Overall mean



◆ Steps in One-Way ANOVA:

1. State hypotheses:

- H_0 : All means are equal
- H_1 : At least one mean is different

2. Calculate:

- **Group means**
- **Overall mean**

3. Compute:

- **SSB (Sum of Squares Between groups)**
- **SSW (Sum of Squares Within groups)**

4. Compute Mean Squares:

- $MSB = \frac{SSB}{df_{between}}$
- $MSW = \frac{SSW}{df_{within}}$

5. Calculate F-statistic

6. Compare with F- or p-value



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F-Statistic (Core of ANOVA)



F-Statistic (Core of ANOVA)

The **F-statistic** is the ratio of variance between groups to variance within groups.

$$F = \frac{MSB}{MSW}$$

Where:

- **MSB** = Mean Square Between groups
- **MSW** = Mean Square Within groups

◆ Interpretation:

- **$F \approx 1$** → No significant difference (accept H_0)
- **$F \gg 1$** → Significant difference exists (reject H_0)



Example of tatic Calculation

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- Suppose:



Degrees of Freedom (Important)

- Between groups:

$$df_{between} = k - 1$$

$$df_{within} = N - k$$

Where:

- k = number of groups
- N = total observations

Key Assumptions of ANOVA

1. Samples are **independent**
2. Data is **normally distributed**
3. **Equal variance** (homogeneity of variance)



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✓ Final Conclusion (Exam-Oriented)



Explain the Resampling approach for Chi-Square test with an example.

Resampling Approach for Chi-Square Test (Exam-Oriented Answer)

1. Introduction

The **Chi-Square test** is used to check whether there is a significant difference between **observed frequencies** and **expected frequencies**.

The **resampling approach** (like **Permutation Test** or **Bootstrap**) is used when:

- Sample size is small
- Distribution assumptions are doubtful
- We want a **data-driven p-value** instead of relying on Chi-square tables

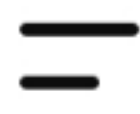
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Resampling involves repeatedly drawing samples from the observed data and



• We want a data-driven p-value
relying on Chi-square tables



2. What is Resampling?

Resampling involves repeatedly drawing samples from the observed data and recalculating the test statistic to build an **empirical distribution**.

Types:

- **Permutation (Randomization) Test**
- **Bootstrap Method**

3. Steps in Resampling Approach (Permutation for Chi-Square)

1. State Hypothesis

- H_0 : No association between variables (independence)
- H_1 : Association exists



2



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3. Resample Data



3. Steps in Resampling Approach (Permutation for Chi-Square)

1. State Hypothesis

- H_0 : No association between variables (independence)
- H_1 : Association exists

2. Compute Observed Chi-Square Statistic

3. Resample Data

- Shuffle one variable randomly (break association)
- Keep marginal totals fixed

4. Recalculate Chi-Square

- Compute statistic for shuffled data

5. Repeat

- Perform 1000+ iterations

6. Calculate p-value

- Proportion of sampled Chi-square $\geq$ observed Chi-square



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Calculate p-value

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Proportion of resampled Chi-square
observed Chi-square

4. Example

Suppose we test whether **Gender** and **Preference (Tea/Coffee)** are independent:

	Tea	Coffee	Total
Male	30	20	50
Female	20	30	50
Total	50	50	100

Step 1: Compute Expected Values

Expected frequency formula:

$$E = \frac{(\text{Row Total} \times \text{Column Total})}{\text{Grand Total}}$$



- Male-Tea = $\frac{50 \times 50}{100} = 25$



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- Female-Tea = 25

=



Step 1: Compute Expected Values

Expected frequency formula:

$$E = \frac{(Row\ Total \times Column\ Total)}{Grand\ Total}$$

- Male-Tea = $\frac{50 \times 50}{100} = 25$
- Male-Coffee = 25
- Female-Tea = 25
- Female-Coffee = 25

Step 2: Compute Chi-Square Statistic

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

$$\chi^2 = \frac{(30-25)^2}{25} + \frac{(20-25)^2}{25} + \frac{(20-25)^2}{25} + \frac{(30-25)^2}{25}$$

$$= 1 + 1 + 1 + 1 = 4$$



Step



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- Shuffle labels (Male/Female) randomly

- Create new tables

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Step 4: Calculate p-value

$$p = \frac{\text{Number of times } \chi^2_{\text{resampled}} \geq 4}{\text{Total simulations}}$$

If:

- 50 out of 1000 ≥ 4
- $p = 0.05$

5. Interpretation

- If $p < 0.05 \rightarrow$ Reject $H_0 \rightarrow$ Association exists
- If $p \geq 0.05 \rightarrow$ Fail to reject H_0



6. Advantages of Resampling Approach

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- Works well with small datasets



$p < 0.05 \rightarrow \text{Reject } H_0 \rightarrow \text{Assumption exists}$



- If $p \geq 0.05 \rightarrow \text{Fail to reject } H_0$

6. Advantages of Resampling Approach

- No strict distribution assumptions
- Works well with small datasets
- More accurate p-values
- Flexible and data-driven

7. Disadvantages

- Computationally intensive
- Requires many iterations
- Slight randomness in results



8. Conclusion



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The resampling approach provides an alternative way to compute p-values using simulated data rather than theoretical tables, making it powerful for